WORKED SOLUTION

STEP I 1991, Question 7

Pure • Separable Differential Equations

STEP I 1991, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q7

The key idea

Both rates are constant multiples of $(1+V/V_0)^2$, so the natural variable is $u = 1+V/V_0$, which satisfies $\frac{\mathrm{d}u}{\mathrm{d}t} = \pm c\,u^2$. Separating the variables shows that the *reciprocal* $w = 1/u$ simply changes linearly: it falls at rate a in a growth year and rises at rate b in a decline year. Everything then reduces to tracking one straight-line graph and asking two questions: does w hit 0 (the money becomes infinite), or does it hit 1 (the Treasury is empty)?

Solution

Setting up

Measure t in years and write $V = kV_0$, so $u = 1+k$ and $\mathrm{d}V/\mathrm{d}t = V_0\,\mathrm{d}u/\mathrm{d}t$. In a growth year

$$V_0\frac{\mathrm{d}u}{\mathrm{d}t} = aV_0u^2 \quad\Longrightarrow\quad \frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}t} = a \quad\Longrightarrow\quad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{u}\right) = -a,$$

and in the same way, in a decline year (while $V > 0$), $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{u}\right) = +b$. So with

$$w = \frac{1}{1+V/V_0},$$

w decreases linearly at rate a during growth and increases linearly at rate b during decline. Note that $V \geq 0$ corresponds to $0 < w \leq 1$: the Treasury is empty exactly when $w = 1$, and $V \to \infty$ exactly when $w \to 0^+$.

The meaning of the inequalities

At the start $w = w_0 = \frac{1}{1+k_0}$, and during the first year $w = w_0 - at$. Then:

- $k_0 \geq 0$ simply says the Treasury cannot hold a negative amount of money.

- The money stays finite throughout the first year exactly when w stays positive up to $t = 1$, i.e. $w_0 - a > 0$, which is

$$\frac{1}{1+k_0} > a \iff 1 + k_0 < \frac{1}{a} \iff k_0 < a^{-1} - 1.$$

  If this failed, V would become infinite at time $t = w_0/a \leq 1$, within the very first year, which is absurd for any model of a real treasury.

Together the inequalities also force $a^{-1} - 1 > 0$, that is $a < 1$; we shall use this at the end.

Surviving the second year

After the first year $w_1 = \frac{1}{1+k_1}$ (and indeed $w_1 = w_0 - a$). During the decline year $w = w_1 + bt$, as long as $w < 1$. The Treasury is still non-empty at the end of the year exactly when w has not reached 1 by $t = 1$:

$$\frac{1}{1+k_1} + b < 1 \iff b < \frac{k_1}{1+k_1}.$$

In particular we need $b < 1$, and then this rearranges to

$$k_1 > \frac{b}{1-b}.$$

If instead $b \geq \frac{k_1}{1+k_1}$, the money runs out at time $t = (1 - w_1)/b \leq 1$ and stays at zero for the rest of the year.

Unlimited growth in the third year

> **Ask yourself** What if the Treasury empties in year two? Could it still blow up in year three?

If the Treasury is empty at the end of year two, year three starts with $w = 1$ and $w = 1 - at$, which stays positive throughout the year because $a < 1$. So an empty Treasury cannot blow up in year three, and we only need to consider the case where money survives year two.

In that case $w_2 = w_1 + b = \frac{1}{1+k_0} - a + b$, and in year three $w = w_2 - at$. The money becomes unlimited during the third year exactly when w reaches 0 by $t = 1$, i.e. $w_2 \leq a$:

$$\frac{1}{1+k_0} - a + b \leq a \iff b \leq 2a - \frac{1}{1+k_0}.$$

We must check that this is consistent with "but not before".

- No blow-up in year one is guaranteed by $k_0 < a^{-1} - 1$, and none can happen in year two, when the money is decreasing.

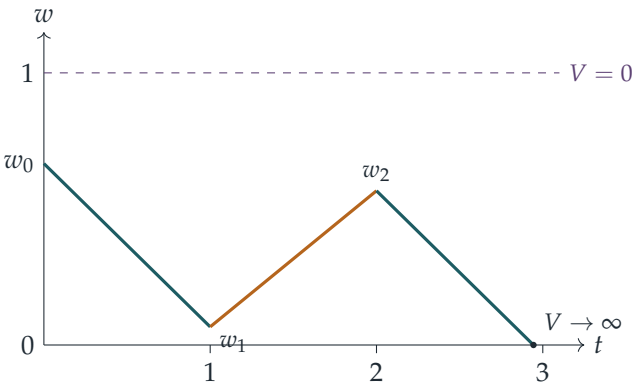
- Money survives year two: if $w_2 \le a$ then $w_2 < 1$ since $a < 1$, so the survival condition $w_2 < 1$ is automatic.

Hence the model predicts unlimited growth in the third year, but not before, exactly when

$$b \le 2a - \frac{1}{1+k_0}.$$

With strict inequality V becomes infinite at time $t = w_2/a < 1$ into the third year; in the equality case $V \to \infty$ just as the third year ends. (Since $b > 0$, this can only happen if $2a > \frac{1}{1+k_0}$, that is, $k_0 > \frac{1}{2a} - 1$.)

The figure shows w against t for $a = 0.6$, $b = 0.5$, $k_0 = 0.5$, a case where the condition holds.



Checks

In the example above, $k_1 = 14$ and the survival condition $14 > \frac{0.5}{0.5} = 1$ holds; then $k_2 = \frac{13}{17} \approx 0.765$, and the third year's blow-up happens at $t = w_2/a \approx 0.944$, consistent with $2a - \frac{1}{1+k_0} - b = \frac{1}{30} > 0$. We also integrated the differential equations numerically (fourth-order Runge–Kutta) for 300 random choices of a, b and k_0: the closed forms for k_1 and k_2 agreed to about 10^{-11}, and both the survival and blow-up criteria predicted the numerical behaviour correctly in every case.

Common slips

- Forgetting that the Treasury can empty in year two. Here it does no harm, because $a < 1$, but it needs to be said.
- Getting the blow-up condition backwards: blow-up corresponds to $1/(1+k)$ reaching 0, not k reaching 0.
- Ignoring the requirement $b < 1$ when rearranging $\frac{1}{1+k_1} + b < 1$ into a condition on k_1.

Beyond STEP

The equation $\dot{u} = au^2$ is the standard example of *finite-time blow-up*: unlike exponential growth, a solution can become infinite at a finite time, so solutions of differential equations need not exist for all time. In a first ODE course this is where the distinction between local and global existence of solutions appears. Boyce and DiPrima, *Elementary Differential Equations*, discusses this when it treats nonlinear first-order equations.

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